NutriGuard

Scan. Know. Eat Safe.

MVP Product Document

iOS App · May 2026 · Confidential

1. Target Audience & Problem Statement

Hundreds of millions of people live with conditions like diabetes, gastritis, high blood pressure, high cholesterol, GERD, and gout. Every single meal is a question they can't confidently answer: is this food okay for me to eat?

Calorie apps were built for gym-goers, not patients. A diabetic doesn't need to know how many calories are in their rice — they need to know if it's going to spike their blood sugar. A gastritis sufferer doesn't care about macros — they want to know if that fried egg on the plate is going to hurt later. No app answers that.

Who This App Is For

Category	Description
Primary User	Adults managing one or more chronic conditions — diabetes, gastritis, high blood pressure, GERD, high cholesterol, gout, celiac, lactose intolerance, and more
Platform	iOS (iPhone) — MVP
Age Range	25–65 years old
Core Pain Point	No app that checks a real meal against a real person's real conditions. Generic apps don't cut it.

The Core Problem

- Calorie apps were made for healthy people — not for people managing a medical condition
- Nobody has built an app that looks at your actual meal and checks it against your actual conditions
- People rely on memory, guesswork, and old advice from their doctor
- Eating the wrong thing over and over makes conditions worse — and nobody's helping them avoid that
- The real question isn't "how many calories?" — it's "can I eat this without paying for it later?"

Feasibility Check

This is buildable right now. GPT-4o Vision can identify food from a photo with solid accuracy. Each scan is one API call at roughly $0.003. The logic is simple. The two main risks are medical liability (handled with disclaimers and Health & Fitness categorisation on the App Store) and AI accuracy (handled by letting users correct the identified food items).

2. Competitor Analysis

There are apps in the food scanning space, but none of them actually solve what NutriGuard is here for.

App	Focus	Condition Support	Key Weakness
Cal AI	Calorie tracking via photo	None	Purely macro-focused. Zero condition awareness.
MyFitnessPal	Calorie & macro logging	None	Built for fitness. No meal photo AI.
GoCoCo	Diabetes food guide	Diabetes only	One condition only. No multi-condition support.
SNAQ	Glucose impact tracking	Diabetes/glucose	Needs a CGM sensor. No photo recognition.
Yuka	Ingredient & additive scan	None	Barcode-only — can't scan a cooked meal.
Fig	Allergen & dietary guide	Allergens, FODMAP	Packaged food only. Not for chronic conditions.

Good Ideas Worth Taking

- Cal AI: Scanning the ingredients panel on packaged food — quick win to add
- Cal AI: Animated streak badges — one of the simplest and most effective ways to keep people coming back
- Yuka: Suggest a healthier swap when an item is flagged — just one extra field in the AI prompt
- Fig: Lead with what's safe, not just what to avoid — positivity goes a long way

3. Branding & Identity

Attribute	Detail
App Name	NutriGuard
Tagline	"Your food. Your conditions. Your answer."

Tone	Warm and straight-talking. Like a friend who happens to know nutrition — not a doctor reading from a chart.
Vibe	Food-inspired and welcoming. Warm oranges, fresh greens, golden yellows. Feels like a kitchen, not a clinic.
Platform	iOS only (MVP)

Brand Personality

- Straight with you — tells you what's actually in your food, even if it's not what you want to hear
- In your corner — built around your specific conditions, not a one-size-fits-all answer
- Human — talks like a person, not a medical textbook
- Simple — takes complicated nutrition science and makes it a single clear verdict

4. Main Features

These are the features people download NutriGuard for. The core value of the app.

Feature 1 — Snap, Type, and Get Your Answer

The user photographs their meal, photographs the ingredients label on a packaged food, or just types what they're eating. NutriGuard sends that to GPT-4o alongside the user's conditions, weight, and age — and gets back a plain-English breakdown of every food item on the plate.

Four ways to scan:

Scan Type	What Happens	What the User Sees
Meal Photo	AI spots every food item on the plate	A verdict for each item, with the option to tap for more info or hit "Eat Anyway"
Ingredients Label	User photographs the nutrition or ingredients panel on a food package	AI reads the label and checks each ingredient against the user's conditions. Same verdict flow as a meal photo.
Type It In	User types what they're eating — e.g. "grilled salmon with rice and steamed veg"	Same full analysis as a photo scan. All verdicts, advice, dish safety score, and meal insight included. Always free.
Unrecognised	AI can't make out what's in the photo	"We couldn't work out what's in this photo — try getting a bit closer or better lighting!"

Verdict System

Every food item gets a score from 0 to 5. That maps to one of five plain-English verdict labels shown as a coloured chip right next to the food name — no numbers, no jargon. Items that are totally fine are shown as green "All Good!" chips. Flagged items get their own coloured chip and an "Eat Anyway" button.

Score	Verdict Chip	Chip Colour	What It Means
5	**All Good!**	Green	Nothing to worry about — enjoy it
4	**Mostly Fine**	Light Green	Barely any impact — totally fine to eat regularly
3	**Eat Less**	Amber	A bit of a concern — keep the portion small and don't make it a habit
2	**Not Ideal**	Orange	Likely to aggravate your condition — worth thinking twice
0 – 1	**Skip It**	Red	Real risk for your specific condition — best to leave this one

What the AI Sends Back — Per Item

- Name — what the food item is
- Score — 0 to 5
- Verdict chip label — one of the five above
- Reasoning — one sentence on why it's flagged (shown when the user taps the item)
- Damage control — 2 sentences of practical advice on what to do if the user eats it. If there's genuinely nothing helpful to say, the AI returns: "We couldn't find anything to offset this — better to skip it next time."

What the AI Sends Back — Whole Meal

- Dish safety score — a single percentage (0–100%) for the entire meal. How safe is this plate for your specific conditions? Shown big and bold at the top of the results screen.
- Meal insight — one observation about the overall meal, shown at the top of the results screen above the item list — the first thing the user reads

Feature 2 — Eat Anyway & Damage Control

Tapping "Eat Anyway" on a flagged item opens the Damage Control page. This is what makes NutriGuard different — it's not here to tell you off. It knows people don't always eat perfectly, and it helps you limit the damage when you don't.

Damage Control Page

- Opens when the user taps "Eat Anyway" on a flagged item
- Shows 2 sentences of practical, condition-specific advice for that item
- Back button returns to the results screen
- "Next" button skips ahead to the next flagged item they haven't looked at yet
- Once visited, the "Eat Anyway" button on the results screen switches to "See Damage Control" so the user can always come back to the advice
- No tracking, no confirmation button — it's purely there to help

Examples of damage control advice

- Diabetes: "Take a 15-minute walk after this meal — it blunts the glucose spike and keeps your blood sugar lower for up to 3 hours."
- High blood pressure: "Have a banana or some avocado on the side — potassium directly cancels out some of the sodium's effect on blood pressure."
- Gastritis: "Follow this with chamomile or ginger tea — it helps calm the stomach lining down."

5. Features That Keep People Coming Back

Lightweight additions that make NutriGuard a daily habit without a lot of extra build time.

Scan Streak

A simple counter that goes up every day the user does at least one scan. Shown on the home screen with a flame icon. Miss a day and it resets. At 8pm on any day with no scan, the app nudges the user with a notification before their streak breaks. It's small, but streaks are surprisingly powerful at building habits.

Share Your Result

After any scan, the user can share a clean verdict card via the iOS share sheet. It shows the dish safety score and a quick summary — great for word-of-mouth. "My lunch was 78% safe for my conditions — NutriGuard." Takes about half a day to build.

6. App Pages

NutriGuard has 3 main tabs plus a scan flow triggered from the home screen. Login and the paywall are standalone screens outside the tab bar.

Login & Sign Up

- Email and password
- Apple Sign-In — one tap, no friction on iOS

Tab 1 — Home

What's on screen

- Scan streak counter with flame icon at the top — motivates the user to keep the habit going
- "Snap a Photo" — opens the camera to photograph a meal or ingredients label
- "Pick from Gallery" — pulls an existing photo from the camera roll
- "Type It In" — opens a text field for a text-based scan (always free)

Scan Flow (triggered from Home)

Photo or Gallery

- Full-screen camera with a single big shutter button
- Gallery picker for choosing an existing shot

Type It In

- Clean text field with a helpful placeholder: "What are you eating? e.g. grilled chicken, mashed potato, steamed broccoli"
- Submit sends the description straight to the AI alongside the user's conditions, weight, and age

Loading (both paths)

- Animated screen with friendly copy: "Hold on, checking your meal... □"
- Sends the input + conditions + weight + age to GPT-4o
- Gets back a structured breakdown: per-item verdicts, scores, reasoning, damage control, overall dish safety score, and a meal insight
- Once done, the user lands on the Final Scan tab so the result is front and centre

Results Screen

The header changes depending on what type of scan it was — so it always feels relevant, not robotic:

Scan Type	Header	Vibe
Meal Photo	"Here's what's on your plate □"	Warm, like you're checking it out together
Ingredients Label	"Let's see what's inside □"	Helpful — reading the label so you don't have to
Type It In	"Here's the breakdown □"	Clear and straight to the point
Unrecognised	"Hmm, we couldn't quite make that out □"	Friendly — not an error message, just a nudge to try again

What's on the results screen

- Dish safety score shown big at the top — e.g. "78% safe for your conditions"
- "All Good!" items shown as green chips — scroll past them quickly if everything's fine
- Flagged items shown as cards with their coloured verdict chip and an "Eat Anyway" button
- Tap any item to open a bottom sheet with the AI's one-sentence explanation
- Meal insight shown at the top — one useful observation about the whole meal, before the "Safe to enjoy" and "Worth a closer look" sections
- Scan is saved to Firestore as soon as this screen loads

Damage Control Page

What's on screen

- Opens when the user taps "Eat Anyway" on a flagged item
- Two sentences of practical advice tailored to this item and the user's conditions
- Back button to return to the results screen
- "Next" button to jump to the next flagged item
- After visiting, the "Eat Anyway" button back on the results screen relabels to "See Damage Control"

Tab 2 — Last Scan

Shows the results of the most recent scan, always. When a new scan finishes, the user lands here automatically. It stays until the next scan replaces it — a permanent home for the last result so the user can always refer back.

Tab 3 — Profile

What's on screen

- Weight (kg) and height (cm)
- Birthday — age is auto-calculated and sent to the AI on every scan
- Conditions — add or remove; preset chips for common conditions (Diabetes, Gastritis, Hypertension, GERD, High cholesterol, Gout, Celiac, Lactose intolerance) plus a free-text input field to add any custom condition not in the list
- Subscription status with an upgrade or manage button
- Sign out

Paywall Screen

Design notes

- Pops up when a free user has used all 3 photo/label scans, or taps a Pro feature
- Warm, food-inspired design. Two plans shown side by side — annual highlighted as best value (saves 58%).
- Lead with the pain: "Unlimited scans — no more running out." Annual plan shown as the sensible choice.

7. Free vs Pro

Free users get 3 lifetime scans for Meal Photo and Ingredients Label. Type It In is always free — no scan limit, no paywall.

Feature	Free	Pro
Meal photo scan	3 lifetime	Unlimited
Ingredients label scan	3 lifetime	Unlimited
Type It In scan	Unlimited	Unlimited
Verdict chips (All Good! / Mostly Fine / Eat Less / Not Ideal / Skip It)	Yes	Yes
Dish safety percentage	Yes	Yes
Damage control advice	Yes	Yes
Eat Anyway flow	Yes	Yes
Scan streak	Yes	Yes
Share result card	Yes	Yes

8. Subscription Pricing

Plan	Price	Equivalent	Notes
Free	**$0**	—	3 lifetime photo/label scans + unlimited text scans. Enough to feel the value.
Pro Monthly	**$9.99 / month**	$0.33 / day	Managed via RevenueCat. Free until $2.5K MRR.
Pro Annual	**$49.99 / year**	$4.17 / month	58% cheaper than monthly. Push this on the paywall.

Unit Economics

- Photo and text scan cost: ~$0.003 per scan (single GPT-4o call)
- At 1,000 Pro users: MRR = $9,990. AI costs ~$300/mo (assuming 3 scans/user/day). Gross margin ~97%.
- Break-even: roughly 50 Pro subscribers covers all infrastructure costs

9. Notifications

All notifications are local — scheduled on the device. No server needed, no Firebase Cloud Messaging. Every message should feel like it's meant for that specific person, not a generic blast.

Trigger	When	Example
Streak at risk	Locally scheduled at 8pm if no scan that day	"Your 5-day streak ends tonight — scan your dinner to keep it alive □"
Streak milestone	When streak hits 7, 14, or 30 days	"7 days in a row — you're actually building a habit here □"
First scan nudge	24 hours after sign-up if still no scan	"You haven't tried your first scan yet — takes about 10 seconds □"

Appendix — Data Architecture

Firestore Structure

Scans are stored as a single document per user (currentScan). There is no diary or historical log — the app keeps only the most recent scan result.

Collection / Field	Type	Notes
`users/{userId}`		
`conditions`	`Array<String>`	Condition names the user selected
`weight`	`Number`	kg — sent to AI on every scan
`height`	`Number`	cm
`birthday`	`Date`	Age calculated client-side, sent to AI
`streakCount`	`Number`	Consecutive days with at least one scan
`lastScanDate`	`Date`	Used for streak logic
`plan`	`String`	free | pro
`lifetimePhotoScansUsed`	`Number`	Free users only — max 3 (photo + label scans)
`users/{userId}/currentScan`		
`scanType`	`String`	meal | ingredients | text | unrecognised
`items`	`Array`	Full AI response per food item
`dishSafetyPct`	`Number`	0–100 — overall dish safety score from AI
`insight`	`String`	AI meal-level insight text
`createdAt`	`Date`	Set when the results screen loads

Disclaimer — Show on Every Results Screen

Every screen showing a food verdict must display this. Required for App Store approval and legal cover.

"NutriGuard gives general wellness info based on widely accepted dietary guidelines. This isn't medical advice — always check with your doctor before changing what you eat."

- User must acknowledge during onboarding before they can proceed
- Small footer disclaimer visible on every verdict and results screen
- Include in the App Store description and privacy policy
- Submit under Health & Fitness, not Medical

— End of Document —